



GE Medical Systems

gemedical.com

Technical Publication

Direction 2318669-100

Revision 1

**GE Medical Systems
On-Site Mobile Setup Procedure**

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ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

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X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

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Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment.

GE Healthcare Services is required to notify the applicable U.S. state agency PRIOR to any source service event involving pin source handling. See NUC/PET Radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	10/03/01	Initial Release of Document
1	9/23/05	Chapter 2 Configuration: Updated "Reconfig the OC". Chapter 3 Functional System Test: Updated test procedure. Chapter 5 Filming: Deleted "Teleradiology (Framegrabber Type) Systems & Aux. Monitors".

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Chapter 1

Start Up Procedure

Section 1.0

Line Input Conditions

The system is shipped with the PDU configured for 480VAC in the USA and for 400VAC in Europe. These are the recommended power sources for this system. If changes are necessary, removing the PDU from its mount is required. This is a 2-man job (minimum).

If the PDU *is* to be reconfigured, use the following procedure:

- 1.) Remove metal cover over right A3 panel.
- 2.) Monitor the No Load Line to Line Voltage at A3 L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 3.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 1-1](#) and [Figure 1-1](#) for tap locations).

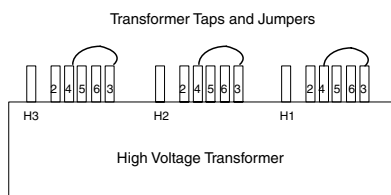


Figure 1-1 PDU Tap Positions (Rear)

Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.

- 4.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 1-1](#).
- 5.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at A3 L1, L2 & L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

- 6.) Replace the metal cover over the right A3 panel.

NO LOAD Line to Line Voltages		TAP CONNECTIONS (All 3 phases must have same the configuration)		
480V*	442 to 518*	3-4*	3-4*	3-4*
400V	368 to 432	2-5	2-5	2-5
* Factory Default				

Table 1-1 PDU 2269902 Line Tap Connections

Section 2.0

Start-Up Procedure (Powerware 9150)

Use this procedure to bring the system up the first time power is applied after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off and turn PDU Circuit Breaker (CB) OFF. This keeps the system computers from needlessly booting up until proper UPS system operation is verified. Note: It is strongly recommended to have the User's Guide for the particular UPS been used.

Check the following before proceeding to power up the UPS unit:

- 480V main power is OFF. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the procedure.
 - Verify that the CT System is ready to receive power and all personnel are clear of the machine.
- 1.) Verify UPS battery circuit breaker, CB1 (on front panel) is in the "OFF" state. Turn off if necessary.
 - 2.) Turn UPS "OFF" via On-Off toggle switch on front control panel (under plastic cover)
 - 3.) Turn the Maintenance Bypass switch (on front panel) fully clockwise to the "BYPASS" position.
 - 4.) Turn "ON" utility power at the main A1 panel.
 - 5.) Press "START" on the auxiliary UPS Control Panel (mounted under or beside the MAIN A1 disconnect).
 - 6.) Turn "ON" the UPS battery circuit breaker, CB1.
 - 7.) Turn the Maintenance Bypass Switch a quarter turn counter-clockwise to the "SERVICE" position.
 - 8.) Start the UPS by turning ON the UPS ON/OFF switch.
 - 9.) The UPS checks its internal functions, synchronizes to utility, performs input voltage and frequency checks, and supplies power to the output. The UPS starts in approximately three to five minutes. The UPS ON, LINE ON, LOAD, and BYPASSED indicators illuminate.
 - 10.) If the UPS does not start and an LED other than the UPS ON indicator is blinking, the UPS is in Automatic Configuration Mode. See section 11.1.2.1.
 - 11.) If the UPS does not start and any of the LED's comes ON, check the fuses in the Disconnect and Control Panels.
 - 12.) If the ALARM LED is On or Blinking (11 short beeps and 5 long beeps), there is an input phase error. The input phases must be rotated at the PDU Terminal Block.
 - 13.) Turn the Maintenance Bypass Switch fully counter-clockwise to the "UPS" position (normal position).

 **CAUTION** Before rotating the Maintenance Bypass switch to the UPS position, verify that the UPS ON, LINE ON, LOAD, and BYPASSED indicators are lit.

- 14.) The BYPASSED indicator turns off when the power is transferred to the inverter, after approximately 30sec.
- 15.) Reinstall the front cover of the UPS.

 **NOTICE** The UPS is now providing 120VAC power to the Table/Gantry and Console. In the event of a momentary or longer 480VAC power loss, the UPS will provide power to the Table/Gantry & Console for up to 10 minutes. YOU CANNOT SCAN with the loss of 480VAC power. The UPS will "beep" occasionally, letting you know that it is on BATTERY BACKUP. When 480VAC power is restored, the UPS and CT system will return to normal operation.

Section 3.0

Configuring Voltage and Frequency

The UPS automatically attempts to match the voltage and frequency with the existing utility. If the UPS is unsuccessful or if you want to bypass the automatic configuration, the voltage and frequency can be selected manually.

3.1 Automatic Configuration Mode

Automatic configuration is performed in two situations: at initial startup or when the Maintenance Bypass switch is in the SERVICE position during UPS startup. The UPS attempts to match voltage and frequency to the existing utility.

When the input voltage and frequency has been determined, the UPS beeps twice and sequences through all the LEDs. One of the LEDs remains blinking. The blinking LED shows the configuration, which has been automatically selected (see the configuration table in section 11.1.2.2).

If the automatic selection is the same as the current configuration, the LED blinks for only 10 seconds. The sequencing LEDs are displayed again and the UPS starts. This automatic startup can be interrupted by pressing the RESET pushbutton quickly during the 10 seconds. This will change the UPS to the Manual Configuration Mode.

If the automatic selection differs from the current configuration setting, the LED blinks (showing automatic configuration selection) until the RESET pushbutton is pushed.

Push the RESET pushbutton momentarily to store this selection in the UPS. The UPS beeps twice and sequences the LEDs. The LED indicating the selection blinks for 10 seconds, the LEDs sequence once again and the UPS starts. This automatic startup can be interrupted by pressing the RESET pushbutton quickly during the 10 seconds; the UPS changes to the Manual Configuration Mode.

If the RESET pushbutton is pushed longer than three seconds, the UPS beeps twice and goes to Manual Configuration Mode.

3.2 Manual Configuration Mode

To set the nominal output voltage and frequency from the front panel:

- 1.) Press and hold the RESET pushbutton while turning on the UPS ON/OFF switch (the | position). Continue to press the RESET pushbutton until the UPS beeps twice and then release.

The UPS indicates Configuration Mode by cycling through the LEDs. One of the LEDs remains blinking. This shows the current configuration according to the following table.

LED	Input/Output Voltage	Output Frequency
ALARM	120/208V	50 Hz
SERVICE	110/220V	50 Hz
OVERTEMP	127/220V	50 Hz
OVERLOAD	100/200V	50 Hz
80%	120/240V	50 Hz
60%	120/208V	60 Hz

Table 1-2 Blinking Configuration

LED	Input/Output Voltage	Output Frequency
40%	110/220V	60 Hz
LOAD	127/220V	60 Hz
BYPASSED	100/200V	60 Hz
ON BATTERY	120/240V	60 Hz

Table 1-2 Blinking Configuration

- 2.) To change a setting, press the RESET pushbutton quickly. The next LED starts blinking. Keep pressing the RESET pushbutton until the desired selection blinks.
- 3.) When the correct setting is blinking, press the RESET pushbutton until the alarm beeps twice. This takes approximately three seconds. The UPS indicates successful configuration by sequencing all the LED's.

Note: The procedures described in section 10.1.2 are part of the POWERWARE 9150 User's Guide.

Section 4.0 Shut-Down Procedure

Use this procedure to bring the system up the first time power is applied after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off and turn PDU Circuit Breaker (CB) OFF. This keeps the system computers from needlessly booting up until proper UPS system operation is verified. Note: It is strongly recommended to have the User's Guide for the particular UPS been used.

Check the following before proceeding to power up the UPS unit:

- 480V main power is OFF. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the procedure.
- Verify that the CT System is ready to receive power and all personnel are clear of the machine.

PROCEDURE

To shut down the UPS do the following:

- 1.) Remove the front panel.
- 2.) Turn the Maintenance Bypass switch to the SERVICE or BYPASS position.
- 3.) Turn "OFF" the UPS ON/OFF toggle switch (under the plastic cover).
- 4.) Turn the UPS battery circuit breaker CB1 and CB2 to the "OFF" position.

Comment: The UPS stops supplying power and is disconnected internally from the batteries. The System is now power by the PDU.

Section 5.0 Powerware 9150 and Plus

5.1 System Off Procedure

- 1.) Depress the "STOP" button on either the UPS Control Panel or A1 Main Disconnect Panel OR depress any room System Emergency Off (SEO) button.
- 2.) Set the manual disconnect switch or circuit breaker in the A1 Main Disconnect to the "OFF"

position.

- 3.) These actions will remove 380-480 VAC main power and UPS power from the CT system and turn off the UPS.

5.2 System ON Procedure - Powerware 9150 UPS

- 1.) Turn the UPS "OFF" via the On-Off toggle switch on its front control panel (under plastic cover).
- 2.) Remove the UPS front cover panel and turn the Maintenance Bypass switch to the "Service" position. Set circuit breakers CB1 & CB2 to "ON".
- 3.) Verify that the CT System is ready to receive power and all personnel are clear of the machine.
- 4.) Set the manual disconnect switch or circuit breaker in the A1 Main Disconnect to the "ON" position.
- 5.) Press the "START" button on the UPS Control Panel. Verify the "UPS ON" light turns on and the system begins to power up.
- 6.) Start the UPS by setting the UPS On-Off toggle switch to the "ON" position.
- 7.) The UPS will begin its internal startup sequence and synchronize to the power line. It will beep periodically while it completes internal startup. After about 3 to 5 minutes it should display the UPS ON, LINE ON, BYPASSED, and LOAD indicators. (If the UPS does not start or indicates any alarm condition, refer to the UPS Operators' manual for additional information.)
- 8.) Turn the UPS Maintenance Bypass Switch fully counter-clockwise to the "UPS" position. The BYPASSED indicator should go out after approximately 30 seconds.
- 9.) Re-install the UPS front cover panel.

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Chapter 2

Configuration

Section 1.0

System Configuration Data Sheets

1.1 Requirement

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their Mobile LightSpeed scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 2-1 Manual Film Composer Options, on page 23](#)
- [Table 2-2 System Network Configuration, on page 23](#)
- [Table 2-3 Networking Application \(Image transfer\) Configuration, on page 24](#)
- [Table 2-4 DASM Laser Camera Configuration, on page 24](#)
- [Table 2-5 DICOM Print Camera Configuration, on page 25](#)
- [Table 2-6 DICOM Print Camera Advanced Configuration, on page 25](#)

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1.2 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS

Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 2-1 Manual Film Composer Options

1.3 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 2-2 System Network Configuration

1.4 Host Ethernet Address

_____ : _____ : _____ : _____ : _____

1.5 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 2-3 Networking Application (Image transfer) Configuration

1.6 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 2-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION	
Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 2-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION	
Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 2-6 DICOM Print Camera Advanced Configuration

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Section 2.0

Reconfig the OC

Tools Required: None

Note: The document collector box that arrived with your system contains the *Software Installation Procedures* manual, which documents the reconfiguration procedure in more detail.

2.1 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWHHMMSS
```

Where

YYYYMMDDWWHHMMSS is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

2.2 Procedure

- 1.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the Install utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the Install Utility Window as shown in [Figure 2-1](#).

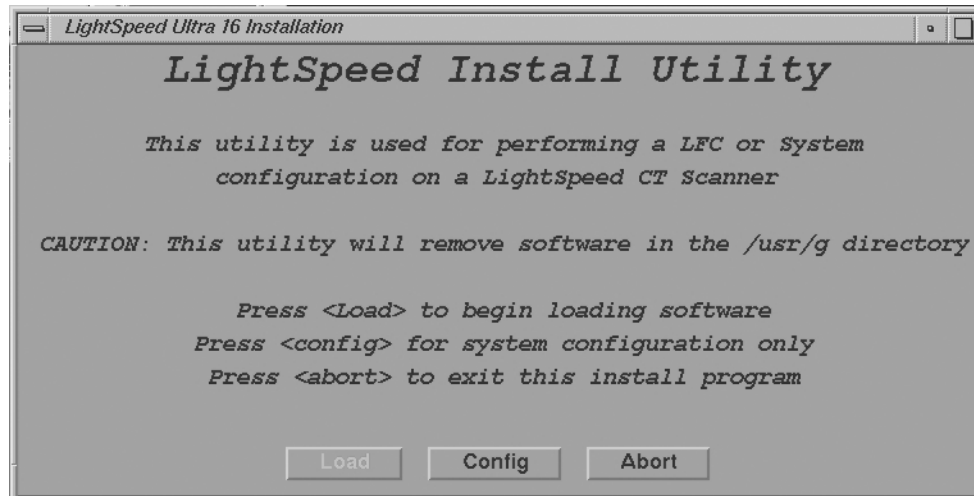


Figure 2-1 Install Utility Window

- 3.) Click on the CONFIG button.

The OC displays the System Configuration - System Settings Screen as shown in [Figure 2-2](#).

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

Figure 2-2 System Settings Screen

4.) Configure System Settings

- a.) Hospital Name (Figure 2-2, item 1) configures the name that will show up on images produced by this scanner. *Example: ST MARYS HOSPITAL*
- b.) Service ID (Figure 2-2, item 2) is issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.

Note:

Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
 - e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: 1
 - f.) Next Diagnostic Exam # *Customer Selected.*
 - g.) Regenerate Database
 - h.) Mobile System Select to tell the software if this CT is in a mobile environment or not.
- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 2-3.
- 6.) Regenerate Database (refer to Figure 2-3) determines whether the Scan Database will be regenerated during reconfiguration. You should select NO.

Note:

If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

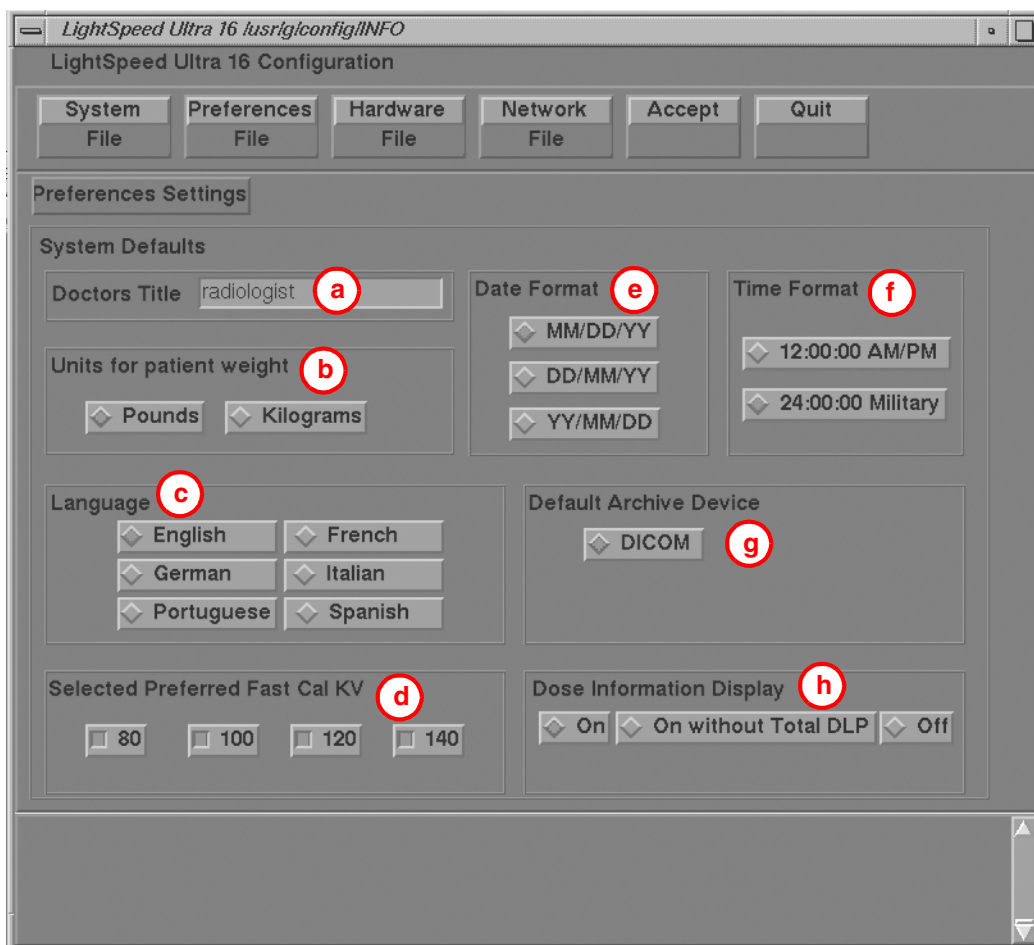


Figure 2-3 Preferences Setup Screen

7.) Configure Preferences Settings

- a.) **Doctors Title** Enter the title for the Doctor. (eg. radiologist)
- b.) **Units for Patient Weight** Tells the software whether pounds or kilograms are being used.
- c.) **Language** configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGUESE, SPANISH).
- d.) **Preferred Fast Cal KV** configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.
- e.) **Date Format** configure the format in which the date will be displayed on the images.
- f.) **Time Format** configure the format in which the time will be displayed on the images
- g.) **Select the site preferred Dose Information Display** option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)
- h.) **Dose Information Display** selects

Comment:

These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

- 8.) Select the **HARDWARE** button to display the Hardware Settings Screen (Figure 2-4).

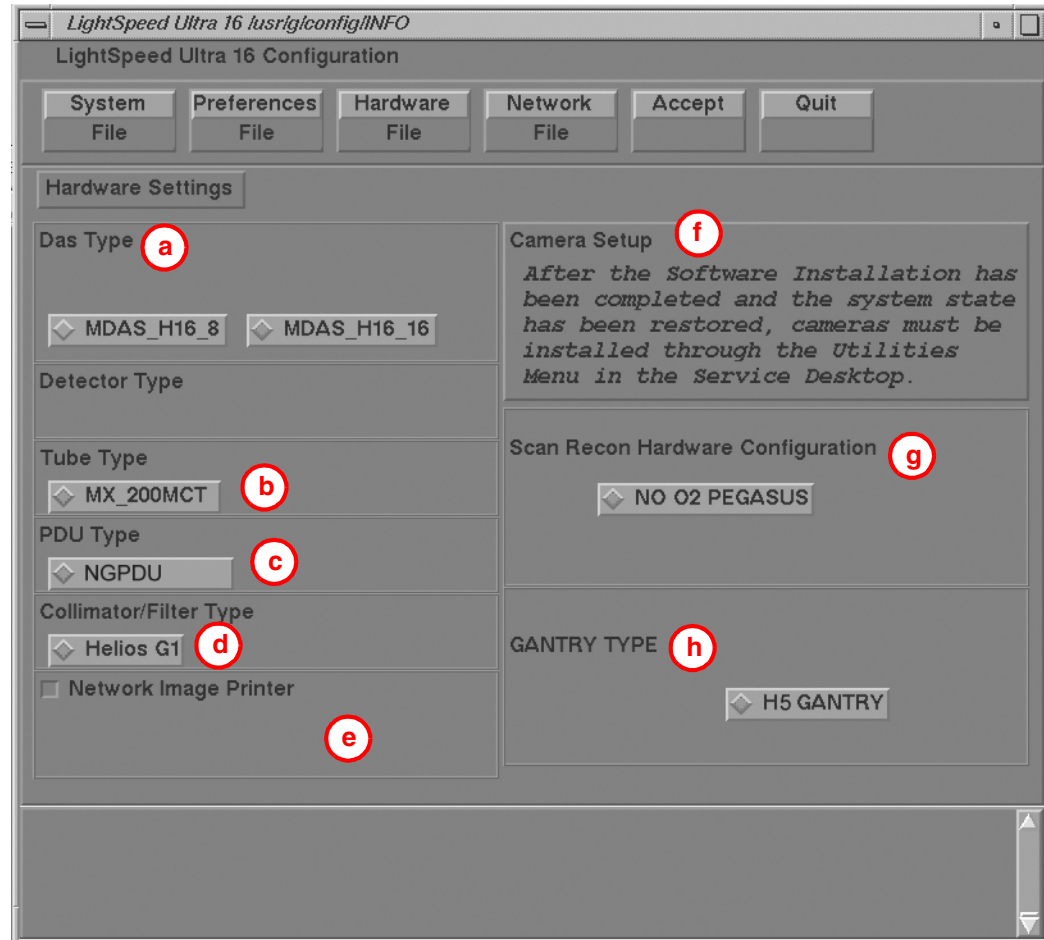


Figure 2-4 Hardware Settings Screen (example only - actual screen may vary)

- 9.) Configure Hardware Settings
- a.) DAS Type configures which DAS the software should expect.
 - b.) Tube Type configures which X-Ray Tube the software should expect.
 - c.) PDU Type configures which PDU the software should expect.
 - d.) Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, LightSpeed 5.X systems only support HELIOS G1.
 - e.) Select Network Image Printer only if the site plans to use a network connected Postscript printer (such as Codonics printers) with this system.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- Comment:**
- f.) Camera Setup:
Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started. See "Filming" on page 47.
 - g.) Scan Recon Hardware Configuration selects whether the O2 is present.
 - h.) Gantry Type selects the type of Gantry connected to this console.

- 10.) Select the **NETWORK** button to display the Network Settings Screen as shown in [Figure 2-5](#).

Figure 2-5 Network Settings Screen

- 11.) Configure Network Settings

Comment: This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

See Chapter 4 for more information and complete details of setting the Hospital/System Network Configuration.

- a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

- b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the CT system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment: The Host Name is typically stmary or ct01.

- c.) Enter the hospital provided IP Address for the system.

- d.) Enter the hospital provided Net Mask (if the CT system is on a subnet).

- e.) Enter the Broadcast Address.

Comment: The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other Sun OS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- f.) Enter the hospital provided Default Gateway IP Address in the `Default Gateway` field (if applicable). If the site network does not use a default gateway, leave the field blank.
- g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:
 - * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital provided `Domain Name` (from the hospital network administrator).
 - * Review all pages to be sure the information entered is correct before proceeding to the next step.

- 12.) Select ACCEPT on the System Configuration Screen.

Comment:

The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named:_____

- 13.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 2-6](#).)



Figure 2-6 Reboot Screen

- 14.) The system will automatically login as `ctuser` after the reboot. Select OK on the Autostart Disabled popup message.

2.3 Applications Start-Up

Open the Console shell window, and type: `st` ENTER. The applications desktop appears on the monitor.

Chapter 3

Functional System Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

1.) Place the QA phantom on the cradle.

- Drive the table to an elevation of 100.
- Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.

3.) Run each series.

Note: **Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.**

4.) Using Diagnostic Data Analysis, review the MSD Plots for each AIR scan.

5.) Complete the scans.

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Chapter 4

Hospital/System Network Integration

Section 1.0

Declaring the System on the Hospital Network

1.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
- 2.) Enter root as a superuser:
Type: `su -` ENTER at the prompt.
Type: `#bigguy` ENTER at the password prompt.
- 3.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the root prompt.
- 4.) Launch the LightSpeed Install Utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the Install Utility Window as shown in [Figure 4-1](#).



Figure 4-1 Install Utility Window

- 5.) Enter the Configuration Routine:
Using the mouse, click on the CONFIG button.
The OC displays the System Configuration - System Settings screen, as shown in [Figure 4-2](#).

Figure 4-2 System Settings Screen

1.2 Configure Network Settings

This screen provides the ability to declare the system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 4-3](#).
- 2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be `<Suite Name>_oc` and the SBC interface will be `<Suite Name>_sbc`.

Figure 4-3 Networks Settings Screen

- 3.) Enter the hospital provided Host Name.
The Host Name identifies the network hostname and AE Title of the system. The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _
- 4.) Enter the hospital provided IP Address.
- 5.) Enter the hospital provided Net Mask (if the system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.
For example:
If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:
 - a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.
 - c.) Enter the hospital provided Domain Name.
- 9.) Record all the Network parameters in the *System Software Load Procedure* Document.

1.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.
The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWHHMMSS.`
(Where `YYYYMMDDWWHHMMSS` is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type `startup` ENTER.

Section 2.0 Declaring Remote Hosts on the System

2.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



2.2 Declaring Advantage NET Remote Hosts on the System

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with Mobile LightSpeed DICOM formats. Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have this system communicating with.

- 1.) Select REMOTE HOSTS, then ADD, from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 4-4](#).

Figure 4-4 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select ADVANTAGE NET as the Network Protocol.
The systems automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

2.3 Declaring DICOM Remote Hosts on the System

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have this system communicating with.

- 1.) Select REMOTE HOSTS, then ADD, from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 4-5](#).

Remote host parameters

Host name

Network address

Network protocol: Advantage NET, DICOM

Port number: 1802

AE Title

Comment

Archive Node: ☐ Auto ☐ Yes ☐ No

Access to local host:

send images? ☐ Yes ☐ No

query/retrieve images? ☐ Yes ☐ No

Custom search? ☐ On ☐ Off

Save Cancel

Figure 4-5 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.

The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.
- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.

Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the System. For most devices, the AE Title is the same as the hostname (Mobile LightSpeed systems are equipped with this feature).

However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).

Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.

- 7.) Select the correct `Archive Node` choice for the device. The `Archive Node` selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - AUTO to have the system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate `Archive Status = Y` to the operator.
 - NO if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct `Access to local host:` settings. These two selections allow you to selectively block the remote host from using the DICOM services as a provider (image **push**-to destination, and a **Query/Retrieve** provider).
 - `Send Images?` Set to YES if the customer wants the system to be able to have images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - `Query/retrieve images?` Set to YES if the customer wants the remote host to be able to review the image database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct `Custom search?` setting. This selection allows the Mobile LightSpeed to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *System Software Load Procedure* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

Section 3.0

Declaring the System on Remote Hosts

3.1 Declaring CT Systems on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the System as an Advantage NET remote host.

3.2 Declaring CT Systems on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the System as a DICOM remote host.

The Mobile LightSpeed System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the Mobile LightSpeed image database.
- A remote host can review the Mobile LightSpeed image database (query) and **pull** selected images (retrieve).

Use the following parameter information to configure the DICOM device/system to either **push** images to the Mobile LightSpeed and/or perform a **Query/Retrieve** operation:

- **Hostname:** Provided by the Hospital Network Administrator. Exactly the same Mobile LightSpeed assigned hostname entered in Network Configuration Screen.
- **Application Entity Title:** Exactly the same entry as the Hostname.
- **Network Address:** Provided by the Hospital Network Administrator. Exactly the same Mobile LightSpeed assigned IP Address entered in Network Configuration Screen.
- **Network Protocol:** DICOM 3.0.
- **Port Number:** For all DICOM service that the System provides, use 4006.
- **Provider Type:** This field concerns the DICOM **Query/Retrieve** provider capability. All systems are study root systems, which allow queries at the exam, series, and image level.
- **Support Worklist:** This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All systems support a filtered search of the image database as part of the DICOM **Query/Retrieve** provider capability.

Section 4.0

Declaring DICOM HIS/RIS Interfaces on the System

4.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

4.2 Loading ConnectPRO Software Option on the System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon. 
- 2.) Click on the UTILITIES icon. 
- 3.) Click on INSTALL.
- 4.) Select INSTALL OPTIONS and click START. The console displays the LightSpeed Software Options window as shown in [Figure 4-6](#).

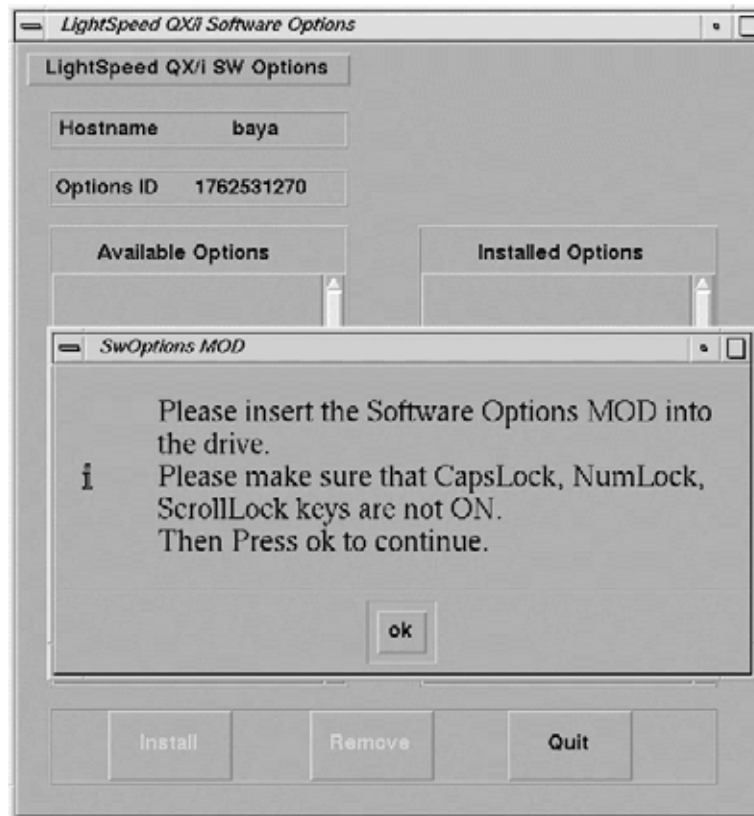


Figure 4-6 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive.
 - 6.) Click on OK to continue.
 - 7.) On the LightSpeed Software Option window, select the CONNECTPRO option in the Available Options window.
 - 8.) Click on the INSTALL button.
- The console may display a message box during the software loading operation. When the system has completed loading the software:
- the Installed Options window displays the ConnectPRO option, and
 - the console displays the ConnectPro Setup window as shown in [Figure 4-7](#).

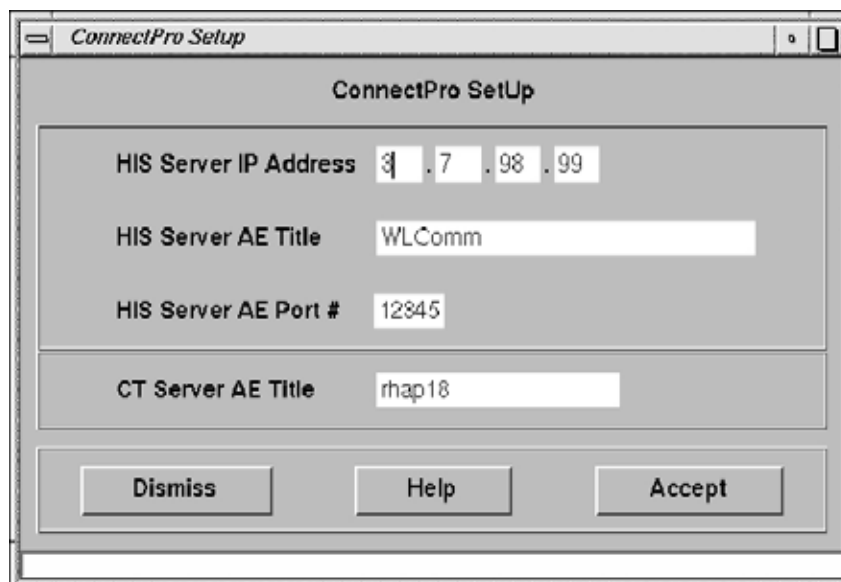


Figure 4-7 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters in the *System Software Installation Procedures* Document.
The system will automatically load the LightSpeed CT Server AE Title (Hostname of the system).
- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the LightSpeed Software Option window shown in [Figure 4-6](#).
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the MOD from the MOD drive, and write protect the side containing the ConnectPRO option.

Chapter 5

Filming

Section 1.0

Declaring DICOM Filming Devices on the System

1.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

1.2 Declaring DICOM Filming Devices on the System



NOTICE Potential For Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.

- 1.) Click on the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 5-2). First, you must click OK in the warning message box. See Figure 5-1.

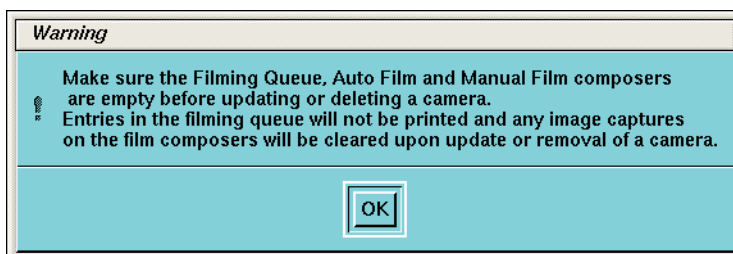


Figure 5-1 Warning Screen

- 4.) Now you are asked a series of questions.
 - a.) To add a new camera, click the ADD button (See Figure 5-2).
 - b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 5-3). If a DASM is present and has not been detected, reboot the OC and run the camera configuration tool again.

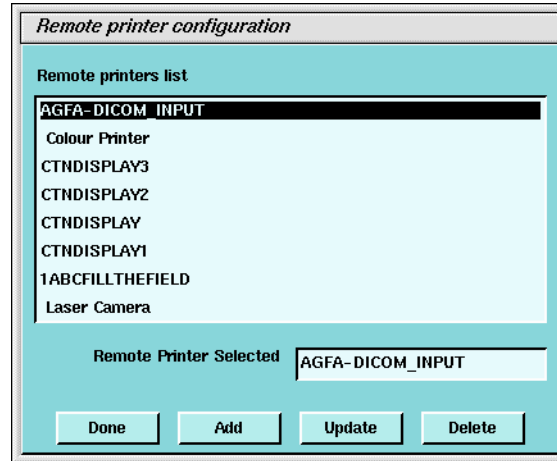


Figure 5-2 Printer Configuration GUI

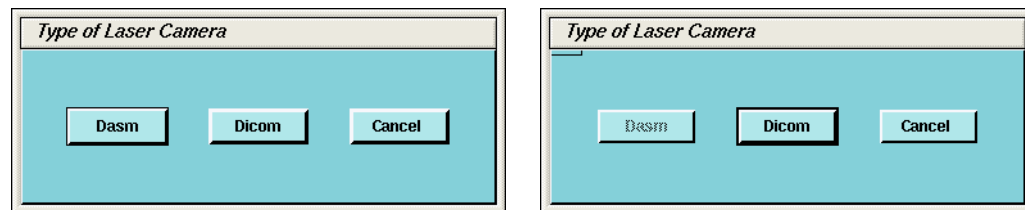


Figure 5-3 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model from the list and click SELECT (See Figure 5-4). Now you must configure it.

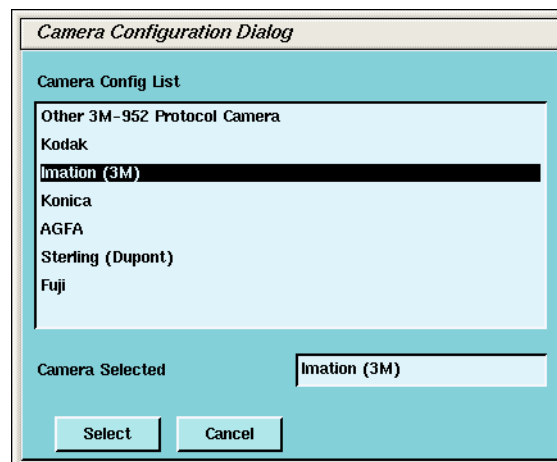


Figure 5-4 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option

only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.

- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH
Dupont/Sterling: SMOOTH
3M/Imation (Laser Camera): SHARP
3M/Imation (Dry View): SMOOTH
Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4). When finished setting parameters, click on DONE and proceed to step 8.

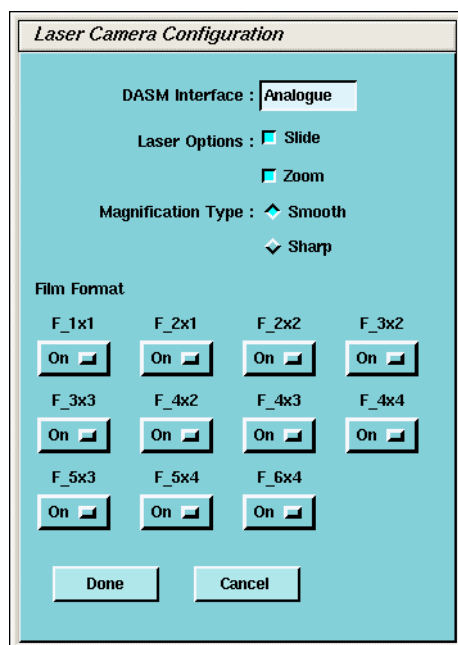


Figure 5-5 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
a.) A list of camera models appears (See Figure 5-6). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

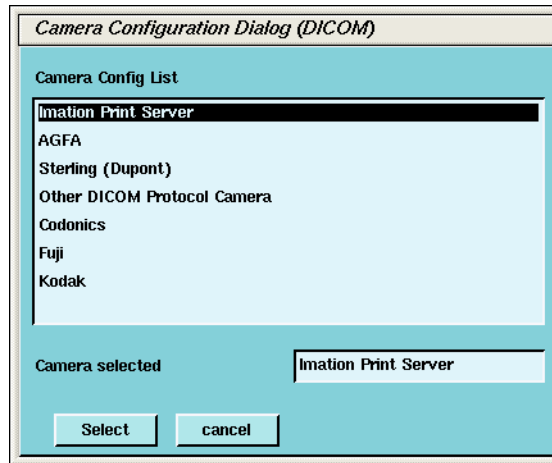


Figure 5-6 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.

b.) Enter the Network Parameters (See Figure 5-7)

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

Figure 5-7 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and CLEAR FILM are supported.
- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - > None No interpolation. This option is not supported by all cameras.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - > Bilinear A 1st order interpolation of pixels. Results in images usually described as blurred. *This algorithm is not normal preferred.*
 - > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 5-7. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).
- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 5-8.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

The screenshot shows a dialog box titled "Camera Attribute Dialog". It contains several sections of controls:

- Image Quality Parameters**:
 - Configuration**: A text field containing "PERCEPTION_LUT=KANAMORI".
 - Maximum Density**: A text field containing "320".
 - Minimum Density**: A text field containing "32".
 - Smoothing Type**: A text field containing "140".
 - Empty Density**: A text field containing "Black".
 - Border Density**: A text field containing "Black".
 - Film Size**: A text field containing "<empty>".
 - Trim**: A text field containing "Yes".
 - Priority**: A text field containing "High".
- Buttons**: "Done" and "cancel" buttons at the bottom.

Figure 5-8 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative. You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) **Configuration** - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: **Recommended Configuration Setting Values:**

Agfa Drystar (MG3000)	PERCEPTION_LUT=KANAMORI	(100)
Imation Dryview (8700)	LUT=0, 7	
Kodak Laser Printer 190	CS434\CN0\PD1.20	

- b.) **Smoothing Type** - Set Smoothing Type to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: **Recommended Smoothing Type Starting Values and Ranges:**

Agfa DryStar (MG3000)	Start Value: 140	Range: 137-150
Imation Dryview (8700)	Start Value: 3	Range: 3-13
Kodak Laser Printer 190	Start Value: Enhanced	Range: Normal

- c.) **Minimum and Maximum Density** - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For **Maximum Density**, input the correct value into the text box. For **Minimum Density**, set it to ON and input the correct value in the text box.

Note: **Recommended Minimum and Maximum Density Starting Values:**

Agfa Drystar (MG3000)	Min.: 20 or 23	Max: 300
Imation Dryview (8700)	Min.: (Blank)	Max: 300
Kodak Laser Printer 190	Min.: 20	Max: 300

- d.) Empty Image Density - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
 - e.) Border Density - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
 - f.) Film Size - Allows the system to specify a particular film size, if selected.
 - g.) Trim - YES produces a white (clear) box surrounding each image.
 - h.) Priority - This sets the print priority.
 - i.) If you have completed entry of advanced parameters, click DONE.
- 8.) **Camera Data Tables:** To locate Install Camera Information: Click on the SERVICE DESKTOP button. On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears. Select each of the cameras that are installed from the list of installed cameras, and click on UPDATE to view the camera's settings. Record the values used to set up each camera in the tables that follow. Extra tables are provided for multiple cameras

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg` where, `name.cfg` is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM CAMERA #1

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 5-1 DASM Camera #1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 5-2 DICOM Camera #1

GUI SETTING	SELECTIONS	VALUES
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 5-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 5-3 DICOM Camera #2

GUI SETTING	SELECTIONS	VALUES
	TCP Listen Port	
	Comments	
Special Set Up	Destination	☐ Magazine ☐ Processor
	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 5-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 5-4 DICOM Camera #3

1.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in `~ctuser/app-defaults/devices` must be manually edited to remove the offending characters in the set configuration line. Invalid characters include `\{ }`

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.
For Trim Off - set TRIM NO
For Trim On - set TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for Mobile LightSpeed (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.
To force a 14x17 film size - set filmSize 14INX17IN

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.
 - To prevent sending the Smoothing Parameter set FB_Smooth FALSE

- | | |
|--|-----------------|
| - To prevent sending the Border Density set | FB_Border FALSE |
| - To prevent sending the Empty Image Density set | FB_EID FALSE |
| - To prevent sending the Minimum Density set | FB_MinD FALSE |
| - To prevent sending the Trim Parameter set | FB_Trim FALSE |

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GEMS Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GEMS Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Chapter 6

Audits & Verifications

Section 1.0 Service Tools

Verify that all service tools are present. A storage cabinet, provided by the unit manufacturer, should contain the following supplied service equipment:

ITEM	SIZE	WEIGHT (TOTAL)
QA Phantom (water filled)	20 x 15 cm	12 lb
20CM Phantom	20 x 7 cm	13 lb
48CM Phantom	48 x 7 cm	25 lb
Phantom Holder	25 x 25 cm	8 lb
Gantry Display	62 x 13 cm	2 lb
FE Box (Purple)	30 x 38 x 30 cm	15 lb
Rear Cover Dollies	158 x 82 cm	25 lb
Front Cover Dollies	85 x 20 cm and 85 x 15 cm	35 lb
Install Support Kit (box)	30 x 30 x 38 cm	20 lb
Tube Hoist Kit	77 x 8 cm and 38 x 15 cm	20 lb
Gantry Control Pad	27 x 20 cm	4 lb

Table 6-1 Equipment to be stored in storage cabinet

Section 2.0 FDO

- Audit and Verify FDO for Completeness
- Call customer center with any discrepancies. FDO paperwork is in the FE box (purple).

Section 3.0

HHS Data Records & Product Locator

3.1

System History (HHS)

This section is to be completed by the Site Field Engineer. All HHS documentation will be included in the FE purple box.

Where Installed _____

City _____ State _____

Customer's Telephone Number _____ Room No. _____

System I.D. No. _____

Operator's Console (Master Control)

Model Number _____ Serial Number _____

System Gantry

Model Number _____ Serial Number _____

Dates of Tests _____ Performed by _____

Table 6-2 Data Record

3.2

Product Locator Cards

Update product locator card for completeness,

Section 4.0

Quality Feedback Form

Complete and return the Quality Feedback Form:

Identification:

GE Van Number

Van Serial Number

Van Manufacturer: _____

GE MAJOR COMPONENTS	MODEL NUMBER	SERIAL NUMBER	LOCATION OR NAME PLATE
Gantry			Rear lower left corner
Table			Non-gantry side of table
Console			Rear lower corner
PDU			Rear lower right corner

Customer Name: _____

Site Address: _____

Van Major Components	Part/Serial Number	Manual Included with the Van?
Air-conditioning	_____	_____
Operator/Maintenance Manual	_____ Rev: _____	_____
Service/Systems Manual	_____ Rev: _____	_____
Unit Identification plaque	_____	_____
Radiation Survey	(Date/ID) _____	_____
Special Installed Options	_____	_____
	_____	_____
	_____	_____
	_____	_____

Are the following requirements met? (If not, comment below.)

Is there a back-up power generator on the van? _____

If present, does the generator provide at least 100 kW? _____

If the generator does not provide at least 100 kW, is the Room Door Interlock held open?

Is there a Transit Warning System installed and connected to the GE System? _____

Is there a labeled CT System main power switch with "lock-out" within 1.5 meters of the PDU?

Is the GE LightSpeed Transit Preparation Procedure clearly posted in the scan-room?

Is there a Service Storage Cabinet in the van for storing Service Dollies, phantoms and tools? _____

Are brackets installed in the van for hanging the gantry side covers during service? _____

Is all required documentation provided (including used van documentation requirements)? _____

For used vans:

Have the Phase I and II Inspection Checklists been filled out and sent to GE? _____

Is the Air Conditioning adequate and operational? _____

Is the radiation shielding adequate? _____ (Attach Radiation Survey or justification.)

Is the van suspension road-worthy? _____ Is there a current DOT Certification? _____

Is the electrical system adequate and documented: _____

Has GE accomplished a Serviceability Review for this configuration? _____

Has GE done a "Final Sign-off" on this van? _____

Cite: Comments, Notes, Problems, Deviations, etc.:

Distribution/filing of this checklist: Completion Date: _____

- 1.) Please file a copy of this check list and all supporting documents at your facility.
- 2.) Please place a copy of this checklist and supporting documents in the van.
- 3.) Fax a copy to:

GE Healthcare
c/o Tom Myers
Fax # 262-312-7369

Section 5.0

Customer Walk-Through

Complete the "Customer Walk-Through" Checklist:

PART #	DESCRIPTION	CHECKED
2115993	Metal-less Cradle Extender	
2115996-2	Metal-less Axial Head Holder	
2115990	Metal-less coronal Head Holder	
2115984	Extender Pad	
46-297592p1	Bag Holder	
46-236664p1 SC 79800	Wrist Strap	
46-297691G1	Arm Board	
46-229452G1	Head/Knee Pads	
46-278968P2	Head/Knee Supports	
46-278986P2	Shin / Ankle Support	
46-229450p1 S/C 40839	Security Strap	
46-222436G1 S/C 37504	6 Chin Straps in plastic bag	
46-297960p1	Foam Insert	
46-222444P1 S/C 37507	Wedge, 10 Degree	
46-222445P1 S/C 37508	Wedge 25 Degree	
46-222921P1 S/C 37983	Sponge	
2115985	Table Pad	
46-266042P1 S/C 09243	Restraint, Wide	
46-266041P1 S/C 09242	Restraint, Narrow	
46-237605P1 S/C 79797	Wrist Strap	
46-297629P1	Strap	
2115997	Axial Head Holder Collector	
46-237412P1	Foam Strap 1PK (3 Pieces)	
46-222443P1	Foam Insert	
46-199388P1 S/C 72944	Insert	
	All Customer Software and Documentation	

Table 6-3 Customer Walk-Through Checklist

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Chapter 7

Install Completion Checklist

Complete the following checklist:

Section 1.0

CT System

- ☐ Functional tests completed.
- ☐ Power and ground audit completed.
- ☐ Reconfigure the database

Section 2.0

CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
- ☐ Teleradiology Connections

Section 3.0

System Cleanup

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

Section 4.0

Install Completion Material Checks

- ☐ Review all HHS documentation.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete system transfer and notify appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.



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